

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Based Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4– Viva Based Questions	Assessment:	Assessment Tool 3 – Viva Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Excellent understanding and accurate application of population, sample, parameters, statistics, and point estimation.. (BL4)		
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Section / Batch:	B. Tech AIML	Date:	21-08-2026

Code of Conduct

I K. lahari (192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Lahari

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. What is statistical inference?

Statistical inference is the process of using sample data to draw conclusions or make predictions about a population.

2. Why is statistical inference important in data science?

It helps data scientists make decisions about a large population using limited sample data and quantify the uncertainty in those decisions.

3. What is the difference between descriptive statistics and inferential statistics?

- **Descriptive statistics:** Summarizes the data we already have.
- **Inferential statistics:** Uses sample data to make conclusions about a larger population.

Example: Average marks of 50 students are descriptive; using those 50 students to estimate the average marks of all students is inferential.

4. What is a population in statistics?

A population is the complete group of individuals or observations that we are interested in studying.

Example: All students in a university.

5. What is a sample?

A sample is a smaller subset selected from the population for analysis.

Example: 200 students selected from the university.

6. Why do data scientists usually work with samples instead of complete populations?

Because studying the entire population can be expensive, time-consuming, or sometimes impossible.

7. What is a population parameter?

A population parameter is a numerical value that describes a population. Examples:

- Population mean μ
- Population proportion P

- Population variance σ^2

8. What is a sample statistic?

A sample statistic is a numerical value calculated from sample data.

Examples:

- Sample mean $\bar{x}$
- Sample proportion $\hat{p}$
- Sample variance s^2

9. Give a real-world example where statistical inference is used in industry.

An e-commerce company can survey a sample of customers to estimate the satisfaction level of all its customers.

10. What are the two major activities involved in statistical inference?

The two major activities are:

1. Estimation
2. Hypothesis testing

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

It is an approach where probability is interpreted as the long-run frequency of events, and population parameters are treated as fixed values.

12. What is the basic idea behind frequentist statistics?

The basic idea is to use repeated-sampling behaviour of data to estimate parameters and test hypotheses.

13. How does the frequentist approach differ from the Bayesian approach?

Frequentist	Bayesian
Parameter is fixed	Parameter is treated as uncertain/random
Uses sample data	Uses data + prior information
Uses p-values and confidence intervals	Uses posterior probabilities and credible intervals

14. In the frequentist approach, what is considered fixed: the parameter or the data?

The population parameter is considered fixed, while the sample data is considered random.

15. What does long-run frequency mean in frequentist statistics?

It means the proportion of times an event occurs when an experiment is repeated a very large number of times.

16. Why is random sampling important in the frequentist approach?

Random sampling helps ensure that the sample is representative of the population and allows valid statistical inference.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

No. In frequentist statistics, the population parameter is considered a fixed but unknown value, not a random variable.

18. Give an example of a frequentist problem in a data science application. A company may test whether a new recommendation algorithm increases the conversion rate compared with the existing algorithm using a hypothesis test and p-value.

C. Point Estimation

19. What is a point estimate?

A point estimate is a single numerical value used to estimate an unknown population parameter.

Example: Using a sample mean of 75 to estimate the population mean.

20. What is an estimator?

An estimator is a statistical rule or formula used to estimate a population parameter.

21. What is the difference between an estimator and an estimate?

- **Estimator:** The formula or method.
- **Estimate:** The numerical result obtained from that formula.

Example:

$$\bar{x} =$$

$$\frac{\sum x}{n}$$

is the estimator, while 78 is the estimate.

22. What is the point estimator for the population mean?

The sample mean:

$$\pi = \frac{\sum x_i}{n}$$

23. How do you estimate a population proportion from a sample?

Use the sample proportion:

$$\hat{p} = x/n$$

where (x) is the number of successes and (n) is the sample size.

24. What is the point estimate of population variance?

The sample variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

25. Why is the sample mean commonly used to estimate the population mean?

Because the sample mean is unbiased, consistent, and efficient under common assumptions.

26. What properties should a good estimator have?

A good estimator should ideally be:

1. Unbiased
2. Consistent
3. Efficient

27. What is an unbiased estimator?

An estimator is unbiased if its expected value equals the true population parameter.

$$E(\hat{\theta}) = \theta$$

28. What is meant by the bias of an estimator?

Bias is the difference between the expected value of an estimator and the true parameter.

$$Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$$

29. What is the difference between bias and variance of an estimator?

- **Bias:** Measures how far the estimator is systematically from the true value.
- **Variance:** Measures how much the estimator changes from sample to sample.

30. Why can two different samples produce different point estimates for the same population parameter?

Because different random samples contain different observations, causing the calculated statistics to vary.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Because it is calculated from a sample rather than the entire population, and different samples can give different results.

32. What is the sampling distribution of an estimator?

It is the probability distribution of an estimator obtained from repeated samples of the same population.

33. What is the standard error?

Standard error measures the variability of a sample statistic or estimator across repeated samples.

For the sample mean: $SE = \frac{\sigma}{\sqrt{n}}$

$$\sqrt{n}$$

34. How is standard error different from standard deviation?

- **Standard deviation:** Measures variability among individual observations.
- **Standard error:** Measures variability of a sample statistic across samples.

35. How does increasing the sample size affect the standard error of the sample mean?

$$SE = \frac{\sigma}{\sqrt{n}}$$

As (n) increases, the standard error decreases.

36. Why does variability in an estimate decrease when the sample size increases?

Larger samples contain more information about the population, so random sampling fluctuations become smaller.

37. What is the relationship between standard error and confidence intervals? The standard error determines how much uncertainty is around the estimate. Generally:

$CI = \text{Estimate} \pm \text{Critical value} \times SE$ A larger SE produces a wider confidence interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect?

No. Different random samples naturally produce different sample means. This is called sampling variability.

E. Confidence Intervals

39. What is a confidence interval?

A confidence interval is a range of plausible values for an unknown population parameter, calculated from sample data.

40. Why is a confidence interval more informative than a point estimate?

A point estimate gives only one value, while a confidence interval gives a range and indicates the uncertainty around the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

If we repeatedly take samples and construct 95% confidence intervals using the same method, approximately 95% of those intervals would contain the true population parameter.

Important: It does not strictly mean there is a 95% probability that the particular calculated interval contains the parameter.

42. What is the difference between a confidence level and a confidence interval?

- **Confidence level:** The percentage such as 90%, 95%, or 99% that describes the reliability of the interval-building procedure.
- **Confidence interval:** The actual numerical range calculated from the sample.

43. What factors determine the width of a confidence interval?

Main factors are:

1. Sample size
2. Data variability
3. Confidence level

Higher variability and higher confidence level make the interval wider, while larger sample size makes it narrower.

44. How does increasing the sample size affect the width of a confidence interval?

Increasing the sample size generally makes the confidence interval narrower because the standard error decreases.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

The interval becomes wider, because we need a larger range to achieve higher confidence.

46. What is the difference between a z-confidence interval and a t-confidence interval?

Z-confidence interval:

- Uses the normal distribution.
- Commonly used when population standard deviation is known or sample size is sufficiently large.

T-confidence interval:

- Uses the t-distribution.
- Commonly used when population standard deviation is unknown, especially for smaller samples.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

When the population standard deviation is unknown, particularly when the sample size is small.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Hypothesis testing is a statistical method used to determine whether there is enough evidence to support or reject a claim about a population.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)? **Null hypothesis (H_0):** Represents the default claim or no significant effect.

Alternative hypothesis (H_1): Represents the claim that we want to find evidence for.

Example:

$$H_0: \mu = 50$$

$$H_1: \mu \neq 50$$

50. How can a confidence interval be used to make a decision about a hypothesis?

Compare the claimed parameter value with the confidence interval. For a two-sided test at 5% significance:

- If the claimed value is inside the 95% CI $\rightarrow$ generally fail to reject H_0 .
- If the claimed value is outside the 95% CI $\rightarrow$ generally reject H_0 .